Enhancing Drug Demand Forecasting & Inventory Management Using Machine Learning

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Group Name: 3Mtt Group C2 Advanced Data Analysis (108)

Problem Statement

1. **Drug Stockouts**
   Pharmacies run out of high-demand drugs, leaving patients without critical medication.

2. **Expiry Waste**
   Overstocked drugs expire before being sold, causing significant financial losses and patient dissatisfaction.

3. **No Forecasting**
   Inventory decisions are made manually with no data-driven demand prediction system.

Why This Matters

- Patient safety is at risk when life-saving drugs are unavailable
- Pharmacies lose revenue when stock expires before sale
- Manual inventory systems cannot scale to hundreds of drugs
- A data-driven system can predict demand and prevent both problems simultaneously

SOLUTION

Project Overview & Solution

This project builds a machine learning system that analyses pharmacy transaction data to predict drug demand, flag stockout risks, and identify drugs about to expire thereby giving pharmacy managers actionable intelligence.

01
Demand Forecasting
To Predict how many units of each drug will be sold using a Random Forest model trained on historical transactions.

02
Stockout Alert System
Classify which drugs will run out of stock within 3 days based on current inventory and predicted sales rate.

03
Expiry Risk Detection
Flag drugs that are likely to expire before being sold, enabling timely discounting or redistribution.

04
Power BI Dashboard
Export clean, model-enriched data to Power BI for an interactive inventory management dashboard.

Tools & Technologies

Data Handling

pandas · NumPy

Loading, cleaning, feature engineering of the 700-row pharmacy dataset

Machine Learning

scikit-learn (Random Forest)

Training regression and classification models for demand and risk prediction

Visualisation

Matplotlib . Seaborn . Plotly

Exploratory Data Analysis

Dashboard

Microsoft Power BI

Interactive inventory management dashboard for pharmacy decision-makers

Environment/platform

Jupyter Notebook · Python 3

Step-by-step analysis workflow combining code, charts, and narrative

Export

CSV · Excel (.xlsx)

Clean data outputs for Power BI ingestion and stakeholder reporting

Methodology & Implementation

1

Data Loading & Inspection

700 pharmacy records loaded. Confirmed zero missing values. Converted dates, checked distributions.

2

Feature Engineering

Created days_to_expiry, stock-status, Year, Month, Days_to_Stockout, expiry risk flag, stockout risk flag etc.

3

Exploratory Data Analysis

Ranked demand by category, trended sales over time.

4

Model Training

Three models trained: Random Forest Regressor (demand), Random Forest Classifier (stockout risk), Random Forest Classifier (expiry risk).

5

Evaluation

Random Forest Regressor scored with MAE & RMSE. Random Forest Classifiers evaluated on Accuracy, Precision, and Recall.

6

Export & Dashboard

Encoded columns removed. Clean CSV exported. Loaded into Power BI for interactive visualisation.

Results & Model Performance

Model 1 — Demand Forecasting (Random Forest Regressor)

13.10
MAE
Mean Absolute Error

15.38
RMSE
Root Mean Sq. Error

The Random Forest model achieved a MAE of 13.1 and a RMSE of 15.4, indicating a moderate level of prediction accuracy. Although the model's average deviation suggests some level of forecasting uncertainty, the predictions are sufficiently reliable to support inventory planning

Model 2 — Stockout Risk Classifier

91%
Accuracy
Stockout Risk

91%
Precision
Stockout Risk

91%
Recall
Stockout Risk

Model 3 — Expiry Risk Classifier

99%
Accuracy
Expiry Risk

99%
Precision
Expiry Risk

99%
Recall
Expiry Risk

The stockout model correctly identifies 91% out of every 100 at risk drugs. The expiry model is near-perfect — it correctly flags 99% of drugs likely to expire before being sold, almost eliminating blind spots.

POWERBI DASHBOARD VISUALIZATION

OVERVIEW

Parameter
- Total Revenue
- Unit Sold

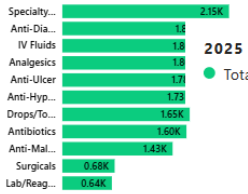
₦9.39M
Total Revenue

17K
Unit Sold

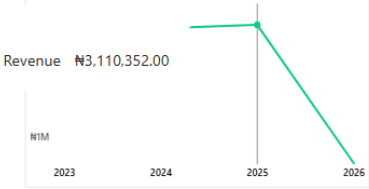
33
Expired Items

100K
Stock_On_Hand

Unit Sold by Category



Total Revenue by Year, Quarter and Month



Drugs with Low Stock but High Demand

Drug_Name	Stock on Hand	Predicted_Demand	High Demand Restock Alert
Oin Morphine 100mg	43	33.58	Urgent Restock
Oin Artemether/Lumefantrine 2g	42	36.87	Urgent Restock
Ear Drops Artesunate 5mg	41	40.28	Urgent Restock
Oin Ranitidine 1g	36	26.27	Urgent Restock
Gutt Adrenaline 250mg	35	33.37	Urgent Restock
Cap Glibenclamide 10mg	34	32.52	Urgent Restock
Oin Gentamicin 5mg	34	23.42	Urgent Restock

EXPIRY RISK DETECTION

Stock_Status
- High
- Low

Expiry Status Flag
All

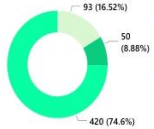
93
High Risk Stockouts

50
Low Stock Count

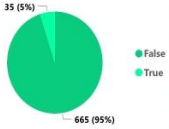
35
Expiry Risk Items

3M
Expiry Loss Exposure

High Risk Vs Low Risk Vs Medium Risk Stockouts



Count of Drug_Name by Expiry_Risk



Expiry Risk Detection

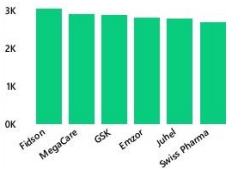
Drug_Name	Stock_Status	Expiry Status Flag
Cap Diazepam 1%	High	Expired
Cap Losartan 250mg	High	Expired
Cap Sulfadoxine/Pyrimethamine 0.5%	High	Expired
Ear Drops Adrenaline 50mg	High	Expired
Ear Drops Hydralazine 100mg	High	Expired
Ear Drops Hydralazine 500mg	High	Close to Expiring
Ear Drops Hydrocortisone 0.5%	High	Expired
Ear Drops Lisinopril 2g	High	Expired
Ear Drops Propofol 10mg	High	Expired
Ear Drops Silver Sulphadiazine 100mg	Low	Expired
Gutt Gentamicin 5mg	High	Expired
Gutt Losartan 100mg	High	Expired
Hepatitis B Strip (2ml)	High	Expired
HIV Kit (Pack of 50)	High	Expired
Inf Chloramphenicol 5mg	High	Expired
Inf Glimepiride 10mg	High	Expired
Inf Glimepiride 2g	High	Expired
Inf Ibuprofen 2g	High	Expired
Inf Ketamine 0.5%	High	Expired
Inf Metformin 50mg	High	Close to Expiring
Inf Tramadol 2g	High	Expired
Inh Amoxicillin 20mg	High	Expired
Inh Atropine 5ml/250mg	High	Close to Expiring
Inh Azithromycin 5ml/250mg	High	Expired
Inh Normal Saline 0.5%	High	Expired
Inj Amlodipine 250mg	High	Close to Expiring

STOCKOUT ALERT SYSTEM

6
Stock Coverage (days)

84
Total Drugs to Order

Sum of Quantity_Sold by Supplier



Stockout Alert System

Drug_Name	Stock_On_Hand	Reorder Point Value	Stock Action
Cap Glibenclamide 10mg	34	110.66	ORDER NOW
Cap Hydrocortisone 500mg	104	120.32	ORDER NOW
Cap Morphine 1g	15	96.50	ORDER NOW
Cap Nifedipine 500mg	114	114.20	ORDER NOW
Cap Omeprazole 250mg	49	115.70	ORDER NOW
Cap Sulfadoxine/Pyrimethamine 0.5%	105	137.84	ORDER NOW
Cap Tramadol 2g	105	117.53	ORDER NOW
Cotton Wool (Size 7.0)	42	75.47	ORDER NOW
Ear Drops Artesunate 5mg	41	133.94	ORDER NOW
Ear Drops Azithromycin 500mg	105	116.63	ORDER NOW
Ear Drops Cimetidine 50mg	121	138.11	ORDER NOW
Ear Drops Gentamicin 20mg	32	82.61	ORDER NOW
Ear Drops Glimepiride 2g	116	137.75	ORDER NOW
Ear Drops Magnesium Trisilicate 500mg	37	74.66	ORDER NOW
Ear Drops Metronidazole Infusion 2g	46	86.33	ORDER NOW
Ear Drops Nifedipine 50mg	40	89.69	ORDER NOW
Ear Drops Nifedipine 5ml/125mg	32	83.15	ORDER NOW
Ear Drops Quinine 5ml/125mg	19	124.70	ORDER NOW
Ear Drops Silver Sulphadiazine 100mg	43	73.10	ORDER NOW
Face Mask (Size 7.0)	5	48.71	ORDER NOW
Giving Set (2ml)	101	114.53	ORDER NOW
Giving Set (Size 7.0)	36	52.67	ORDER NOW
Gutt Adrenaline 250mg	35	113.21	ORDER NOW
Gutt Cimetidine 250mg	106	137.66	ORDER NOW
Gutt Clindamycin 2g	49	94.37	ORDER NOW
Gutt Dextrose Saline 5ml/125mg	113	133.49	ORDER NOW
Gutt Diazepam 10mg	24	132.26	ORDER NOW

Conclusion and Recommendations

The study shows that combining demand forecasting with risk prediction provides a practical system for improving drug inventory management, helping to reduce stockouts and minimize expiry losses.

Recommendations

Pharmacies should use predicted demand together with safety buffers to guide timely restocking decisions

They should also implement continuous monitoring and regularly update the model with new data to maintain accuracy and effectiveness